

Smart i-LAB Testbed

Technical Review: IoT platform, Digital Twin, CV occupancy, and operations

pctiope/smart-i-lab-testbed

May 2026

Review Objective

What this deck documents

- The current combined repository after the Zone 5 mmWave recency update landed on `main`.
- The end-to-end data path from lab devices to TimescaleDB, REST, Digital Twin, and CV/ML consumers.
- Operational boundaries between source deploy, staging Compose, production systemd, retraining, and model delivery.

What is intentionally excluded

- API keys, populated `.env` values, RTSP secrets, MQTT passwords, and private tokens.
- Generated runtime state under `data/`, `model/runs/`, and `logs/`.

System Purpose

Lab platform

- Smart i-LAB Room 308 testbed for sensor acquisition, device control, and remote experimentation.
- MQTT devices and Home Assistant inputs are normalized into TimescaleDB-backed history.
- REST endpoints provide read paths for Digital Twin and model consumers, plus controlled actuator paths.

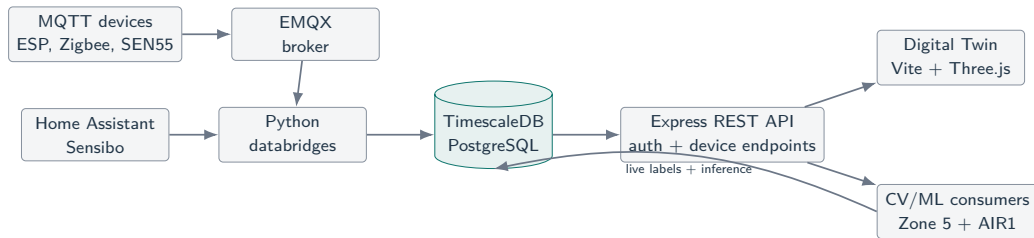
Occupancy intelligence

- Zone 5 package trains a dedicated CNN-1D occupancy model using CV labels plus AIR-1, power, mmWave, SEN55, and time features.
- AIR1 all-zones package trains one shared model over zones 1–15 using two-camera per-zone labels.
- Live dashboards expose model outputs, CV ground truth, health, and video streams.

Repository Map

Path	Role
SSL-IoT1-REST/	Express REST API, OpenAPI files, Dockerfile, auth, device reads, actuator endpoints.
Smart-iLAB-Python-Files/	ESP, Zigbee, and Sensibo databridges from MQTT or Home Assistant into TimescaleDB.
Smart-iLab_DigitalTwin/ migrations/	Vite and Three.js Digital Twin, built and served by nginx in Docker. Versioned schema migrations for core tables, registries, hypertables, and retrofit steps.
zone5_cv_time_features_package/	Zone 5 CV labels, sensor joins, training, promotion, live inference, dashboard, and CI/CD docs.
air1_all_zones_cv_time_features_package/	Shared all-zones AIR1 model, two-camera labels, collector, trainer, and operations app.
docs/, .github/workflows/	IoT1 audit/remediation docs and production/staging workflow automation.

End-To-End Architecture



Schema-as-code

- `migrations/apply.py` applies lexical SQL migrations once and records versions in `schema_migrations`.
- Core tables cover users, transactions, groups, error logs, and device registries.
- Per-device sensor tables are discovered lazily by databridges.

TimescaleDB shape

- Device tables are converted to hypertables when created or retrofitted.
- The current retrofit adds `PRIMARY KEY(timestamp)` where duplicate timestamps do not block it.
- Dirty legacy tables remain an operator cleanup item, with dedupe tooling documented.

REST API And Security Status

Area	Current state	Review note
Auth	API-key checks, access-level validation, query cache.	CV packages send x-api-key; Bearer headers are ignored harmlessly.
SQL safety	User, group, transaction, and ingest paths parameterized or allow-listed.	The historical audit lists 46 findings closed at code level.
Runtime hardening	Helmet, rate limits on auth/write paths, request timeout, body cap, non-root containers.	Behavioral QA still depends on lab infrastructure.
Compatibility	AIR1 and Zone 5 consumer endpoint shapes are unchanged.	Recommended client timeout remains below server cap.

`docs/IOT1_AUDIT.md`, `docs/IOT1_REMEDIATION.md`, and `docs/IOT1_CV_COMPATIBILITY.md` capture the evidence trail.

Role

- Web visualization of lab status and selected controls.
- Demonstrates the REST API as the integration surface for users and downstream tools.
- Built with Vite and bundled Three.js assets.

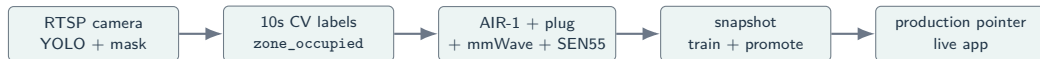
Current packaging

- Multi-stage Docker build: Node builds static assets, nginx serves them.
- API base URL comes from build-time `VITE_API_URL`.
- Content-Security-Policy is present but deliberately permissive pending later tightening.

CV/ML Package Split

Dimension	Zone 5	AIR1 all-zones
Model scope	One model for Zone 5.	One shared model over zones 1–15.
Rows	One row per 10-second timestamp.	One row per timestamp and zone_id.
Labels	YOLO person count aggregated into zone_occupied.	Two-camera mask labels aggregated into occupied.
Inputs	AIR-1, smart plug, mmWave, SEN55, time features.	AIR-1, shared SEN55, missing indicators, time features.
Excluded	Missingness is diagnostic, not a model input.	Power and mmWave are outside the contract.
Serving	Zone 5 dashboard, SSE, history, MJPEG.	All-zones ops view, two MJPEG feeds, per-zone probabilities.

Zone 5 Occupancy Pipeline



- Persistent services: person counter, MQTT aggregator, SEN55 collector, live collector, live app, Vite frontend.
- Trainer timer is separate from collection and uses a retrain lock plus immutable snapshots.
- Promotion atomically updates `model/production_run.txt`; the live app reloads on pointer change.

Zone 5 Feature Contract

Raw signals retained

- AIR-1: temperature, RH, CO2, PM2.5.
- Smart plug: `power_s5`.
- mmWave: `mmwave_s5`.
- SEN55: PM, temperature, humidity, VOC, NOx.
- Time: hour and day-of-week `sin/cos`.

Current contract

- Contract marker:
`zone5_mmwave_recency_v1`.
- Added past-only mmWave recency features: 1m, 3m, 5m occupied fractions and minutes since last occupied.
- Missing indicators remain metadata/diagnostics and are excluded from model inputs.
- Legacy missingness-decoupled artifacts are supported for live loading, but new training writes the recency contract.

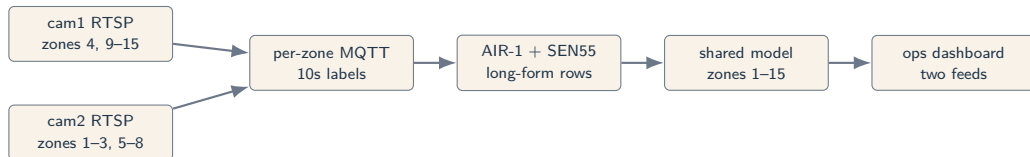
Backend API

- `/api/health`: model, collector, RTSP, and freshness state.
- `/api/current`: latest inference and sensor context.
- `/api/history`: prediction history, CV labels, mmWave lookback context.
- `/api/video.mjpg`: file-backed annotated frame stream.

Frontend behavior

- Vite dashboard displays probability, CV truth, history, MSE, sensor sparklines, and split-pane video.
- New history view surfaces mmWave lookback fraction rather than only the instantaneous bit.
- Generated data, model runs, logs, and runtime environment values remain out of git.

AIR1 All-Zones Pipeline



- Labels are keyed by `timestamp + zone_id`; missing camera coverage stays null, not false zero.
- `zone_id` is retained for grouping, windows, audit, splits, and display, but is not a model input.
- Production inference is gated by `model/production_run.txt`.

Deployment Surfaces

Surface	Purpose	Boundary
Root Compose	IoT1 REST API, Digital Twin, databridges.	External production PG/EMQX/HA by default; dev overlay can colocate PG and EMQX.
Zone 5 systemd	Production source deployment for live services.	User-systemd services, production ports 8000/8015; source deploy does not start trainer.
Zone 5 Compose	Staging deployment path.	Separate branch/path and staging ports 8005/8016.
Model delivery	Promotion of generated model artifacts.	Separate from source deploy; protects generated state and production pointers.
AIR1 systemd/Compose	Parallel package deployment surfaces.	Same collection/training split, no mmWave or power contract import.

Testing And Verification

Code validation

- Python compile checks for Zone 5 modules and web app.
- CV ground-truth unit coverage for contracts, splits, promotion gates, parser behavior, live loader checks, and UI affordances.
- Docker Compose config/build checks in CI/CD paths.

Operations validation

- Health endpoints for live apps and Vite frontend.
- Production pointers: `current_run.txt`, `production_run.txt`, and retrain status.
- Trainer locks avoid overlapping expensive fits.
- IoT1 migration and smoke-test sequence remains operator-gated.

Current Status Snapshot

Area	Status	Marker
IoT1 code remediation	Audit findings closed at code level; deployment cutover pending backup, migration, deploy, smoke test, and log tailing.	operator-gated
Zone 5 source	mmWave recency contract is now on main; package tests and compile checks pass locally.	ready
Zone 5 operations	Production systemd and staging Compose are intentionally separate. Re-training/model delivery remain distinct workflows.	ready
AIR1 all-zones	Shared-model package has persistent service/timer surfaces and preserved null labels for missing camera coverage.	ready
Generated artifacts	Runtime data, logs, model runs, and populated env files stay out of source control.	ready

Risks To Track

Platform risks

- Live IoT1 database migration still requires backup and operator execution.
- Legacy per-device tables with duplicate timestamps need dedupe before every table can receive the timestamp primary key.
- Full behavioral QA requires lab LAN, TimescaleDB, EMQX, Home Assistant, and devices.

Model and runtime risks

- Training quality depends on enough both-class CV label coverage across calendar windows.
- Upstream AIR-1 outages can degrade live feature freshness.
- New contract versions require careful artifact compatibility and dashboard context updates.

Roadmap

1. Complete IoT1 production cutover: backup, migrations, hardened container deploy, admin-key smoke test, and subscriber log watch.
2. Finish table cleanup for duplicate timestamp histories, then re-run the hypertable/primary-key retrofit.
3. Keep Zone 5 manual-first retraining proof before enabling scheduled production retraining.
4. Add monitoring around model freshness, label class balance, promotion outcomes, and upstream API latency.
5. Maintain AIR1 and Zone 5 contract separation: AIR1 stays AIR-1/SEN55/time only; Zone 5 owns power and mmWave.

Review Takeaways

- The repository is a combined lab platform plus two CV/ML occupancy packages; the integration point is the hardened IoT1 REST API over TimescaleDB-backed history.
- Zone 5 now represents mmWave as both an instantaneous signal and past-only recency features, while keeping missingness out of the model input contract.
- Deployment is intentionally split: IoT1 cutover is operator-gated, Zone 5 production stays user-systemd, Compose is staging, and model promotion is separate from source deploy.
- The remaining work is operational discipline: cutover, cleanup, monitoring, and controlled retraining automation.

Topic	Primary source paths
IoT1 architecture and remediation	README.md, docs/STATUS.md, docs/IOT1_REMEDIATION.md
Migrations and schema model	migrations/README.md, migrations/*.sql
Zone 5 pipeline and contract	zone5_cv_time_features_package/PIPELINE.md, zone5/feature_contract.py
AIR1 all-zones contract	air1_all_zones_cv_time_features_package/PIPELINE.md, README.md
Deployment and CI/CD	zone5_cv_time_features_package/CICD_SYSTEMD.md, CICD_DOCKER_COMPOSE.md, .github/workflows/